

## PH.D. QUALIFYING EXAM 2023: PROBABILITY

Rules: This is a four hour exam consisting of two Parts - Part I and Part II, both of which should be answered. You may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN “information sheet” for EACH OF THE TWO COURSES (Stat 592 and Stat 654) on which this exam is based. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Finally, write your answers clearly and legibly in the space provided after each problem, **preferably using pens**, leaving an impression deep enough for the papers to be scanned. Good luck!

**Answer sheets start on page 4.**

## PART I

This part consists of *three exercises* and *one problem*. Each of the three exercises are independent of each other, and independent of the problem. The exercises and the problem may be attempted in any order.

**Exercise 1.** Let  $(X_n)_{n \geq 1}$  be a sequence of random variables. Prove the equivalence

$$X_n \text{ converges in probability to } 0 \iff E[\min(1, |X_n|)] \rightarrow 0.$$

**Exercise 2.** Let  $X$  be a discrete random variable valued in  $\{1, 2, 3, \dots, n\}$  with  $P(X = j) > 0$  for all  $j = 1, \dots, n$ . Prove that for any integer  $k \in \{1, \dots, n-1\}$  and the events  $E = \{X \geq k\}$  and  $F = \{X \geq k+1\}$  we have  $E[X|E] \leq E[X|F]$ .

**Exercise 3.** Consider a random variable  $X$  valued on  $[-1, 1]$  with density  $f$  which is symmetric, that is  $f(x) = f(-x)$  for all  $x \in [-1, 1]$ . Let  $Y$  be a discrete random variable taking only the values 1 and  $-1$  satisfying  $P(Y = 1|X = x) = \frac{1}{1+\exp(-x)}$  for every  $x \in [-1, 1]$ . What is the density of the random variable  $V = YX$ ?

**Problem.** Consider a sequence of iid random variables  $(X_i)_{i \geq 1}$  such that  $P(X_i > 0) = 1$  and  $E[\sqrt{X_i}] < +\infty$  but  $E[X_i] = +\infty$ . Throughout, we use the notation  $\mu = E[\sqrt{X_i}]$ . You may use without proof that the integral  $\int_1^{+\infty} \frac{1}{u^\alpha} du$  is finite for all  $\alpha > 1$ , and that  $\lim_{n \rightarrow +\infty} (1 - \frac{c}{n})^n = \exp(-c)$  for any real  $c$ .

- (1) Applying the law of large numbers to  $\frac{1}{n} \sum_{i=1}^n \sqrt{X_i}$ , prove that  $P(\sum_{i=1}^n X_i \leq (\mu + \varepsilon)^2 n^2) \rightarrow 1$  for any constant  $\varepsilon > 0$  independent of  $n$ .

**In the rest of the problem, the goal is to understand whether a high-probability upper bound much smaller than  $n^2$  can be proved.**

- (2) Consider some fixed constant  $s \in (\frac{1}{2}, 1]$  and iid  $(X_i)_{i \geq 1}$  such that  $P(X_i > u) = u^{-s}$  for all  $u \geq 1$ . What is the range and the density of  $X_i$ ?

**Throughout the remaining questions except the last one, the  $X_i$ 's are now iid with the density derived in part (2).**

- (3) Prove that  $E[\sqrt{X_i}] < +\infty$ .  
 (4) For any real  $y > 1$ , find an expression of  $P(\max_{i=1, \dots, n} X_i > y)$  as a function of  $P(X_1 > y)$  and  $n$  only.  
 (5) What is the limit of  $P(\max_{i=1, \dots, n} X_i > n^{1/s})$  as  $n \rightarrow +\infty$ ?  
 (6) Is it possible, for any sequence of iid random variables  $X_i$  satisfying  $E[\sqrt{X_i}] < +\infty$ , that  $\lim_{n \rightarrow +\infty} P(\sum_{i=1}^n X_i \leq Cn^{1.99}) = 1$  for a constant  $C > 0$  independent of  $n$ ?

## PART II

This part consists of *one exercise* and *one problem*. The exercise and problem are independent. The exercise and the problem may be attempted in any order.

**Exercise.** Let  $U$  be a discrete random variable valued on  $\{1, 2, 3\}$  with pmf given by  $P(U = 1) = \frac{1}{2}$ ,  $P(U = 2) = \frac{1}{3}$ ,  $P(U = 3) = \frac{1}{6}$ .

(a) Construct a coupling  $(X, Y)$  by providing the pmf of  $(X, Y)$  as a three-by-three table such that

- $X$  has the same distribution as  $U$ ,
- $Y$  has pmf  $P(Y = 1) = 0$  and  $P(Y = k) = P(U = k|U \geq 2)$  for each  $k = 2, 3$ .
- $P(X \leq Y) = 1$  and  $P(Y \leq X + 1) = 1$ .

Your answer only needs to provide the table. All conditions above must be satisfied.

(b) Prove or disprove that the coupling obtained in the previous question is an optimal coupling (in the sense that the TV distance between the two marginals equals  $P(X \neq Y)$ ).

**Problem.** Throughout, we consider a finite state space  $\mathcal{X}$ , and irreducible and aperiodic transition matrix  $P$  on  $\mathcal{X}$  and a Markov Chain  $(X_t)_{t \geq 0}$  with transition matrix  $P$ . Denote by  $\mathbf{P}_x[\cdot]$  the probability sign when  $\mathbf{P}[X_0 = x] = 1$  and define the total variation (TV) distance between two probability measures  $\mu(\cdot)$  and  $\nu(\cdot)$  as follows:

$$\|\mu - \nu\|_{TV} = \frac{1}{2} \sum_{x \in \mathcal{X}} |\mu(x) - \nu(x)|.$$

We recall the definitions

$$d(t) = \max_{x \in \mathcal{X}} \|P^t(x, \cdot) - \pi\|_{TV}, \quad \bar{d}(t) = \max_{x \in \mathcal{X}, y \in \mathcal{X}} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{TV}$$

as well as the mixing time

$$t_{mix} = \min\{t \geq 0 : d(t) \leq 1/4\}.$$

Questions:

- (1) Prove that  $\bar{d}(t) \leq 2d(t)$  for all  $t \geq 0$ .
- (2) Prove that  $d(t) \leq \bar{d}(t)$  for all  $t \geq 0$ .
- (3) Prove that  $d(t+1) \leq d(t)$  for all  $t \geq 0$ .
- (4) Prove that  $d(kt_{mix}) \leq \frac{1}{2^k}$  for all integers  $k \geq 2$ . (You may use properties of the distance  $\bar{d}(\cdot)$  from the lectures without proving it, but you need to clearly and rigorously state such properties before using it).
- (5) Let  $T \geq 2$  be an integer. Let  $Z$  be a random variable independent of  $(X_t)_{t \geq 0}$  with uniform distribution on  $\{1, 2, 3, \dots, T\}$ . In words,  $X_Z$  is the value of the Markov Chain after  $Z$  steps, where  $Z$  is random and independent of the chain.  
Express as a sum of  $T$  terms the probability  $\mathbf{P}_x[X_Z = y]$  for each  $y \in \mathcal{X}$ .
- (6) We are interested in the following question: How large should the integer  $T$  be so that  $X_Z$  is close to  $\pi$ .

Prove that if  $T = 7t_{mix}$  then  $\max_{x \in \mathcal{X}} \|\nu_x - \pi\|_{TV} \leq \frac{1}{4}$ , where  $\nu_x$  is the distribution defined as  $\nu_x(y) = \mathbf{P}_x[X_Z = y]$ .

Remarks for question (6): You may split a sum over  $T$  into 7 “bags” of  $t_{mix}$  summands each. You may use without proof that  $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots = 2$ . At this question, you will get 90% of the credit if you prove instead that  $T = Ct_{mix}$  implies  $\max_{x \in \mathcal{X}} \|\nu_x - \pi\|_{TV} \leq \frac{1}{4}$  for a numerical constant  $C$  different than 7, e.g., 8 or 14, provided the numerical constant does not depend on anything.

**ANSWER SHEET PART I**

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(2) Prove that  $d(t) \leq \bar{d}(t)$  for all  $t \geq 0$ .



- (3) Prove that  $d(t+1) \leq d(t)$  for all  $t \geq 0$ .

- (4) Prove that  $d(k t_{mix}) \leq \frac{1}{2^k}$  for all integers  $k \geq 2$ . (You may use properties of the distance  $\bar{d}(\cdot)$  from the lectures without proving it, but you need to clearly and rigorously state such properties before using it).

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